Exercise 2

employees

Id	first_name	last_name	department	salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	52000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Questions:

① Write a SQL query to find the total number of employees in the company

```
SELECT count (id) As total_number_of_employees
FROM employees;
```

total_number_of_employees
10

② Write a SQL query to find the total_salary paid to all employees in the IT department

```
SELECT sum (salary) As total_salary
FROM employees
WHERE department = 'IT';
```

total_salary
220000

③ Write a SQL query to calculate average salary of employees in the HR department

```
SELECT Ava(salary) as average_salary
FROM employees
WHERE department = 'HR';
```

average_salary
49500

④ Write a SQL query to find the highest and lowest salary in the company

```
SELECT MAX(salary) As highest_salary,
       MIN(salary) AS lowest_salary
FROM employees;
```

highest_salary	lowest_salary
62000	48000

⑤ Write a SQL query to group employees by department and display the total salary paid in each department

```
SELECT department,
       sum(salary) AS total_salary
FROM employees
GROUP BY department;
```

department	total_salary
IT	220000
HR	99000
Finance	119000
Marketing	105000

⑥ Write a SQL query to count how many employees work in each city

```
SELECT city,
        Count (id) As number_of_employees
FROM employees
GROUP BY city;
```

City	number_of employees
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

⑦ Write a SQL to group employees by department, calculate average salary in each department, and order the results in descending order of average salary.

```
SELECT department,
        Avg (salary) As average_salary
FROM employees
GROUP BY department
Order by average_salary DESC;
```

department	average_salary
Finance	59500
IT	55000
Marketing	52500
HR	49500

⑧ Write a SQL query to find departments where the total salary paid exceeds 100000

```
SELECT department,
        sum(salary) AS total_salary
From employees
GROUP BY department
HAVING sum(salary) >100000;
```

department	total_salary
IT	220000
Finance	119000
Marketing	105000

⑨ Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order

```
SELECT city,
        count(id) AS number_of_employees
FROM employees
GROUP BY city
HAVING count(id) >1
Order by count(id) DESC;
```

City	number_of_employees
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

(10) Write a SQL query to find the department with the highest average salary

```
SELECT department,
        AVG (salary) AS average_salary
FROM employees
GROUP BY department
ORDER BY AVG (salary) DESC
LIMIT 1;
```

department	average_salary
Finance	59500